

# L30 BAS AI Incident Review Template

## Pre-Irreversibility Civilizational Incident Review | v1.0

L30-BAS codes are reference aids. Writing L30-BAS codes is optional.

A boundary review may be performed by answering the boundary questions without writing the codes.

### Purpose

Purpose: Use this template to review AI incidents under the LUMINA-30 boundary condition. The central question is whether human refusal authority remained effective before irreversible impact.

### Use order

- ☐ Record incident context and scope.
- ☐ Reconstruct the timeline and evidence.
- ☐ Complete L30-BX-01 to L30-BX-04.
- ☐ Complete incident review sections 1-12.
- ☐ Assign L30-CI and record L30-OUT-01.

### Important note

This document is descriptive and non-binding. It does not indicate legal compliance, safety certification, deployment approval, or absence of risk.

Review context

|                            |                            |
|----------------------------|----------------------------|
| Review ID                  | Date / Time                |
| Incident title             | Incident date              |
| Reviewed system            | Reviewer                   |
| Organization               | BAS version / source       |
| Primary user               | Incident type              |
| Severity                   | Impact scope               |
| Responsible scope          | Review scope               |
| Evidence source set        | Record preservation status |
| Reviewer role              | Experience level           |
| Primary user / stakeholder | Review language            |
| Independent human reviewer | Evidence completeness      |

Severity: Low / Medium / High. Impact scope: research environment / institutional systems / infrastructure systems / external environment. Responsible scope: system operator / vendor / institution / oversight body / unknown.

Initial incident summary

# 1. Incident Overview

Record what occurred, how it was detected, and the first observable effect.

|                         |                                                                 |
|-------------------------|-----------------------------------------------------------------|
| What occurred?          | Describe the event in plain operational terms.                  |
| Detection path          | How was the incident first detected?                            |
| Initial system behavior | What behavior was observed before, during, and after detection? |
| Initial impact          | What persons, systems, or environments were affected?           |

Evidence / notes / rationale

## 2. Incident Classification

Classify the incident before evaluating responsibility or outcome.

|                                                                 |                                                                                         |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <input type="checkbox"/> <b>Autonomous research system</b>      | Research agent, experiment automation, model-driven exploration.                        |
| <input type="checkbox"/> <b>Recursive improvement attempt</b>   | Self-improvement, self-modification, or capability feedback loop.                       |
| <input type="checkbox"/> <b>Tool-chain escalation</b>           | Expansion through tools, APIs, agents, plugins, or automation chains.                   |
| <input type="checkbox"/> <b>Infrastructure interaction</b>      | Interaction with power, network, logistics, finance, medical, or institutional systems. |
| <input type="checkbox"/> <b>Autonomous operational behavior</b> | System action outside direct step-by-step human control.                                |
| <input type="checkbox"/> <b>Other</b>                           | Record description and reason for classification.                                       |

**Evidence / notes / rationale**

### 3. System Capability Profile

Record the capability, autonomy, tool access, privilege, and operating envelope of the system.

|                               |                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------|
| Model / system class          | Model type, agent type, automation layer, or hybrid architecture.                |
| Training / update mode        | Static, continuous, reinforcement, self-supervised, hybrid, or unknown.          |
| External tool access          | None / restricted / broad / unknown. Record concrete tools.                      |
| Environment interaction       | Simulation only / digital infrastructure / physical systems / external services. |
| Privilege and execution speed | Permission level, automation speed, deployment cadence.                          |

Evidence / notes / rationale

## 4. Control and Oversight Structure

Identify who could supervise, stop, refuse, delay, or override the system before irreversible impact.

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Human oversight model</b>   | Direct supervision / periodic supervision / post-hoc monitoring / no effective supervision. |
| <b>Intervention capability</b> | Immediate shutdown / delayed intervention / indirect control / no reliable control.         |
| <b>Decision authority</b>      | Human approval required / advisory only / system autonomous / unknown.                      |
| <b>Refusal point</b>           | Where could a human refusal have changed the outcome?                                       |
| <b>Audit visibility</b>        | Were logs, approvals, and system actions independently reviewable?                          |

**Evidence / notes / rationale**

## L30-BAS Boundary Check

Complete this section before drawing final conclusions. If evidence is missing or the review depends on AI-only closed-loop self-evaluation, use Invalid (Not Verifiable).

| Code      | Check question                                                            | Evidence / notes                                                                                    | Result                                                                                                                  |
|-----------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| L30-BX-01 | Was the system still pre-irreversible?                                    | Timeline, deployment state, execution boundary, rollback possibility.                               | <input type="checkbox"/> Satisfied<br><input type="checkbox"/> Not satisfied<br><input type="checkbox"/> Not verifiable |
| L30-BX-02 | Was Human Refusal Authority still effective before irreversible impact?   | Who could stop, refuse, delay, or override? Was that authority practical, not merely formal?        | <input type="checkbox"/> Satisfied<br><input type="checkbox"/> Not satisfied<br><input type="checkbox"/> Not verifiable |
| L30-BX-03 | Is there sufficient evidence to verify the boundary condition?            | Timeline record, refusal / stop record, system log, audit log, independent human review record.     | <input type="checkbox"/> Satisfied<br><input type="checkbox"/> Not satisfied<br><input type="checkbox"/> Not verifiable |
| L30-BX-04 | Can procedural validity be confirmed under LUMINA-30 boundary conditions? | If BX-01 to BX-03 are not satisfied or not verifiable, do not treat refusal authority as preserved. | <input type="checkbox"/> Satisfied<br><input type="checkbox"/> Not satisfied<br><input type="checkbox"/> Not verifiable |

**Boundary evidence notes / L30-OUT-01 draft**

## 5. Pre-Irreversibility Boundary Assessment

Evaluate whether the system approached or crossed a potential irreversibility threshold.

|                                  |                                                                                      |
|----------------------------------|--------------------------------------------------------------------------------------|
| External irreversible impact     | None / potential / observed. Record concrete external effects.                       |
| Loss of human decision authority | No loss / partial degradation / effective loss.                                      |
| Autonomous capability escalation | No escalation / controlled escalation / unbounded escalation.                        |
| Rollback possibility             | Practical rollback / formal-only rollback / no rollback / unknown.                   |
| H > C(t)                         | Was human intervention capacity greater than AI amplification at the relevant point? |

Evidence / notes / rationale



## 6. Institutional Responsibility Structure

Record whether responsibility and control authority remained continuous and identifiable.

|                               |                                                                             |
|-------------------------------|-----------------------------------------------------------------------------|
| Responsible institution       | Which institution had responsibility for the system or process?             |
| Operational control authority | Who had operational control at the relevant time?                           |
| Oversight body                | Was any independent oversight body involved?                                |
| Responsibility continuity     | Clearly defined / partially defined / undefined.                            |
| Responsibility gap            | Where did handoff, outsourcing, automation, or delegation create ambiguity? |

Evidence / notes / rationale

## 7. Intervention Feasibility

Evaluate whether meaningful human intervention remained possible in practice, not merely in form.

|                               |                                                               |
|-------------------------------|---------------------------------------------------------------|
| Stop capability               | Immediate / delayed / ineffective / no mechanism.             |
| Operational containment       | Full / partial / none / unknown.                              |
| Time to intervene             | Was the available time sufficient for a human decision?       |
| Authority to intervene        | Could the relevant person actually stop or delay the system?  |
| Evidence of intervention path | Is there a record showing how intervention would have worked? |

Evidence / notes / rationale

## 8. Incident Outcome

Record outcome, system status, affected scope, and unresolved behavior.

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| System status      | Controlled shutdown / rollback / ongoing investigation / unresolved behavior.           |
| Impact scope       | Research / institutional / infrastructure / external environment.                       |
| Human impact       | Personnel, users, patients, citizens, operators, or affected groups.                    |
| Operational impact | Service disruption, decision chain effect, deployment change, or infrastructure effect. |
| Evidence status    | Complete / partial / missing / contested.                                               |

Evidence / notes / rationale

## 9. Structural Lessons

Identify structural weaknesses revealed by the incident without assigning unsupported blame.

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| Oversight weakness    | What oversight mechanism failed, arrived late, or was absent?  |
| Intervention weakness | Which stop, delay, override, or refusal path was insufficient? |
| Evidence weakness     | Which record was missing, unverifiable, or AI-dependent?       |
| Escalation pathway    | What pathway was not previously identified?                    |
| Repeat risk           | Could the same structure recur elsewhere?                      |

Evidence / notes / rationale

# 10. Post-Incident Stabilization Measures

Record immediate stabilization and evidence preservation measures before normal operation resumes.

|                              |                                                                                  |
|------------------------------|----------------------------------------------------------------------------------|
| Operational suspension       | Was the affected function suspended or restricted?                               |
| System architecture review   | Was architecture, tool access, privilege, or deployment path reviewed?           |
| Oversight restructuring      | Was review authority, approval path, or escalation path changed?                 |
| Independent audit            | Was an independent review initiated?                                             |
| 48-hour stabilization record | What was preserved, frozen, or isolated within the initial stabilization period? |

Evidence / notes / rationale

# 11. Civilizational Boundary Reflection

Examine whether the incident revealed a condition where human judgment, responsibility, or intervention could be structurally bypassed.

|                              |                                                                            |
|------------------------------|----------------------------------------------------------------------------|
| Human judgment bypass        | Could the system or process bypass meaningful human judgment?              |
| Responsibility discontinuity | Was responsibility broken across systems, teams, vendors, or institutions? |
| Intervention impossibility   | Was there a point where intervention became impossible or only symbolic?   |
| Boundary warning             | Did the incident reveal a pre-irreversibility boundary warning?            |
| Future reference             | What should future reviewers check first?                                  |

Evidence / notes / rationale

Final evaluation output: L30-CI

|                                                            |                                                                                                                       |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> L30-CI = Valid                    | Use only when BX-01, BX-02, BX-03, and BX-04 are all satisfied.                                                       |
| <input type="checkbox"/> L30-CI = Invalid                  | Use when a boundary condition is verified as not satisfied.                                                           |
| <input type="checkbox"/> L30-CI = Invalid (Not Verifiable) | Use when evidence is missing, records are insufficient, or the review depends on AI-only closed-loop self-evaluation. |

*Valid means valid only under LUMINA-30 boundary conditions. It does not indicate legal compliance, safety certification, deployment approval, or absence of risk.*

L30-OUT-01 Summary finding

12. Review Classification

|                                                                              |                                                                                         |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <input type="checkbox"/> <input type="checkbox"/> Technical incident         | Technical failure or system malfunction was central.                                    |
| <input type="checkbox"/> <input type="checkbox"/> Governance failure         | Decision, oversight, approval, or responsibility structure failed.                      |
| <input type="checkbox"/> <input type="checkbox"/> Structural oversight gap   | A gap in monitoring, refusal, evidence, or escalation handling was revealed.            |
| <input type="checkbox"/> <input type="checkbox"/> Boundary-condition warning | The incident indicates possible loss of effective human refusal before irreversibility. |

Classification rationale

**Reviewer confirmation**

**Reviewer name**

**Organization**

**Role**

**Review date**

**Signature**

*This template is descriptive and non-binding. BAS version / source should be recorded in the review context section.*